Physical Chemistry CUME, April 6, 2018

Chemical Equilibrium

1. Consider the following reaction:

$$A + B \rightleftarrows P$$

Write down the equilibrium constant of this reaction in terms of the enthalpy and entropy of A, B, and AB at a given temperature T and pressure p. [20 points]

2. Consider the complex reaction described by

$$A \underset{k}{\overset{k}{\rightleftarrows}} B \underset{k}{\overset{k}{\rightleftarrows}} C$$

Suppose that the only molecular species present at time $t = 0$ is A. Find the equilibrium concentrations of A, B, and C in terms of the initial concentration of A. [25 points]

3. The standard Gibbs energy of reaction for the following decomposition reaction is $+118.08\ kJ\ mol^{-1}$ at $2300\ K$:

$$H_2O(g) \rightarrow H_2(g) + \frac{1}{2}\ O_2(g)$$

What is the degree of dissociation of H_2O at $2300\ K$ and 1.00 bar?
Note: The degree of dissociation, α, is defined as the fraction of reactant that has decomposed; if the initial amount of reactant is n and the amount at equilibrium is n_{eq}, then $\alpha = (n - n_{eq})/n$. To simplify calculations, you can assume $\alpha \ll 1$. [25 points]

4. Consider the following reaction:

$$2\ N_2O(g) + 3\ O_2(g) \rightleftarrows 2\ N_2O_4(g)$$

Calculate the equilibrium constant (K_c) at the same conditions that the following equilibrium constants have been determined? [30 points]

$$2\ N_2(g) + O_2(g) \rightleftarrows 2\ N_2O(g), K_c = 1.2\times10^{-35}$$

$$N_2O_4(g) \rightleftarrows 2\ NO_2(g)\ , K_c = 4.6\times10^{-3}$$

$$\frac{1}{2}N_2(g) + O_2(g) \rightleftarrows\ NO_2(g)\ , K_c = 4.1\times10^{-9}$$